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Day 2

Aim: Create an authorized local target lab. Verify connectivity, identify the target IP, run a basic Nmap host/port scan and document discovered services

Objective:

To create an authorized local target lab, verify connectivity between Kali Linux and the target machine, identify the target IP address, perform a basic Nmap host and port scan, and document the discovered open ports and running services.

Requirements:

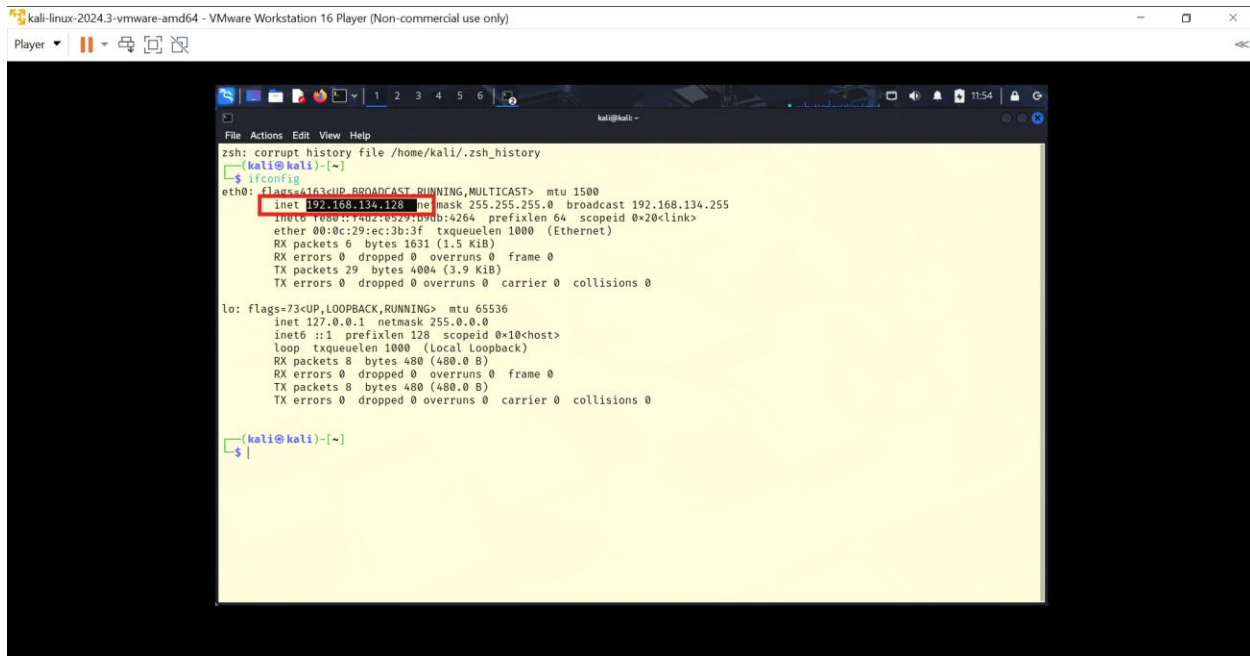
- Kali & Metasploitable 2
- Nmap

Theory:

Nmap is a network scanning tool used to identify active hosts, open ports, and running services on a target system. In this practical, Kali Linux is used as the scanning machine and Metasploitable 2 as the authorized target. Connectivity is first verified using ping, followed by an Nmap scan to discover and document open ports and services.

Practical:

Step 1. Indentify Ip addresses..

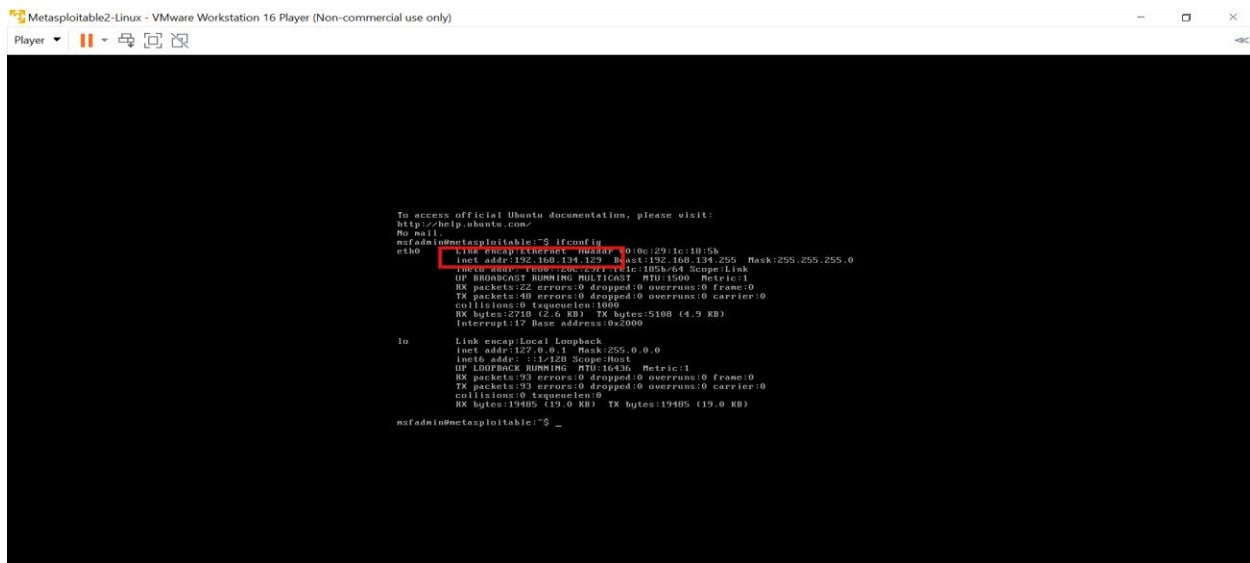


```
kali-linux-2024.3-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)
Player
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.134.128 netmask 255.255.255.0 broadcast 192.168.134.255
    inet6 fe80::f424:527:d9db:4264 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:ec:3b:3f txqueuelen 1000 (Ethernet)
    RX packets 6 bytes 1631 (1.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 29 bytes 4004 (3.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$
```

Figure 1. Kali ip: 192.168.134.128



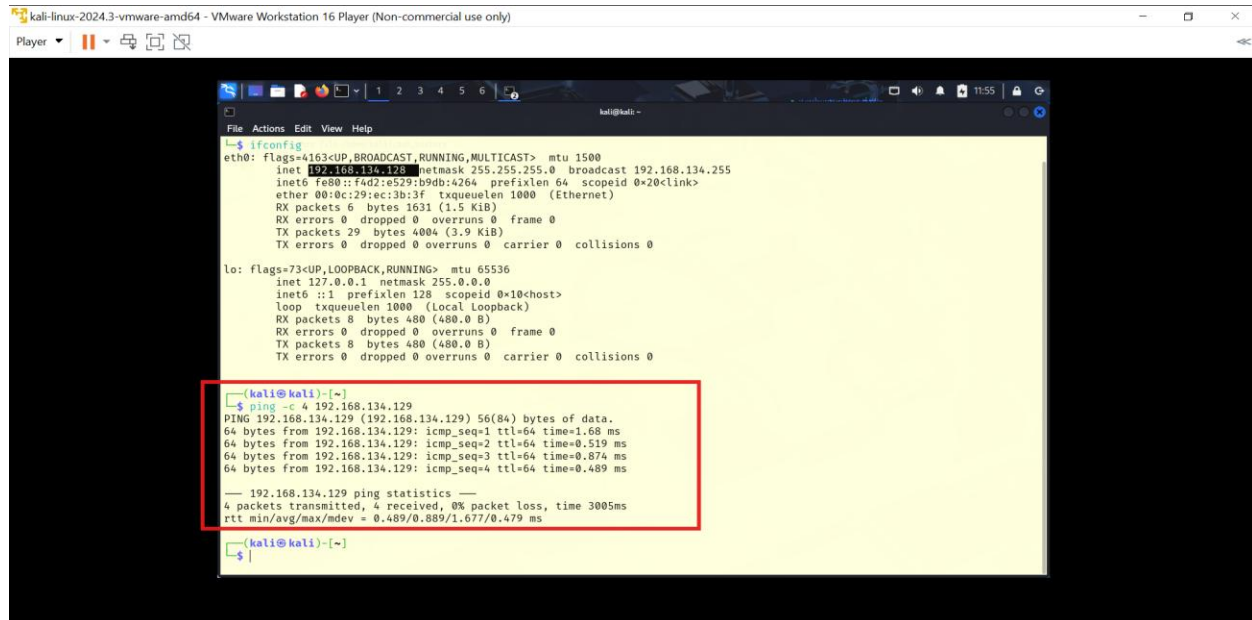
```
Metasploitable2-Linux - VMware Workstation 16 Player (Non-commercial use only)
Player
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail
msfadmin@metasploitable:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet addr:192.168.134.129 Bcast:192.168.134.255 Mask:255.255.255.0
    inet6 fe80::f424:527:d9db:4264 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:ec:3b:3f txqueuelen 1000 (Ethernet)
    RX packets:22 errors:0 dropped:0 overruns:0 frame:0
    TX packets:40 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:1000
    RX bytes:2210 (2.6 KB) TX bytes:5100 (4.9 KB)
    Interrupt:17 Base address:0x2000

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet addr:127.0.0.1 Bcast:127.0.0.1 Mask:255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets:13 errors:0 dropped:0 overruns:0 frame:0
    TX packets:13 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:0
    RX bytes:19405 (19.0 KB) TX bytes:19405 (19.0 KB)

msfadmin@metasploitable:~$
```

Figure 2. Metasploitable ip: 192.168.134.129

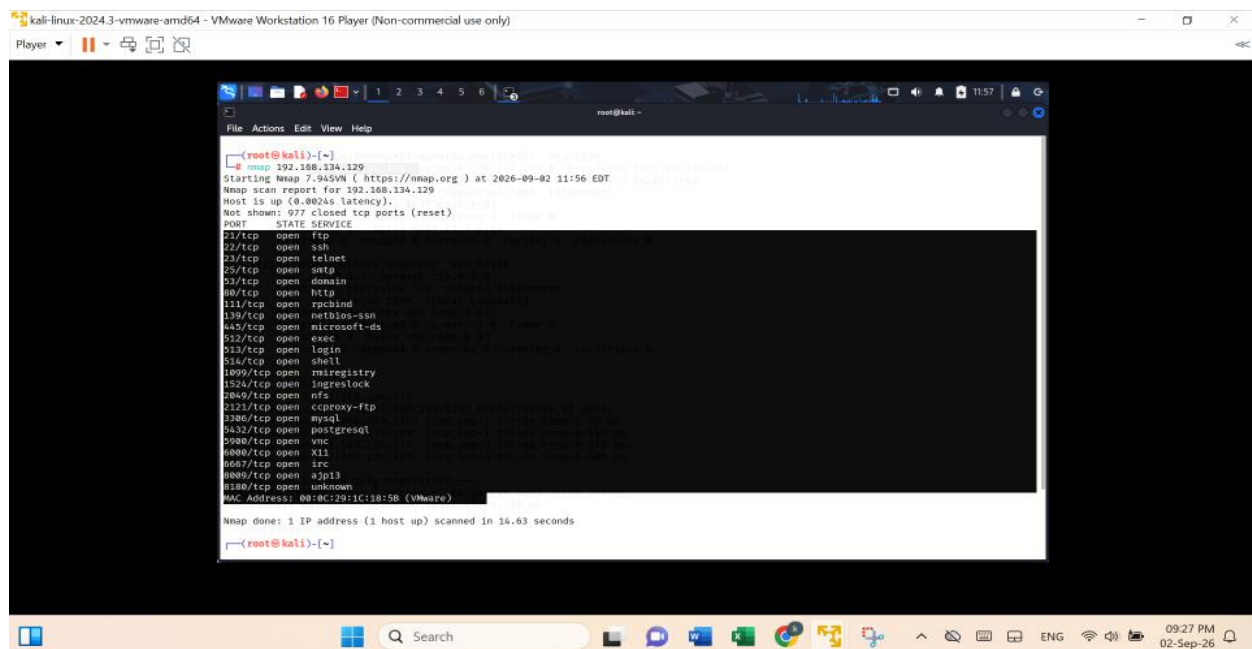
Step 2. Verify Connectivity.... Between kali and metasploitable.



```
kali@kali: ~  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.134.129 netmask 255.255.255.0 broadcast 192.168.134.255  
    inet6 fe80::f4d2:c529:b9db:4264 prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:ec:3b:3f txqueuelen 1000 (Ethernet)  
    RX packets 6 bytes 1631 (1.5 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 29 bytes 4004 (3.9 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 8 bytes 480 (480.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 8 bytes 480 (480.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
kali@kali: ~  
$ ping -c 4 192.168.134.129  
PING 192.168.134.129 (192.168.134.129) 56(84) bytes of data:  
64 bytes from 192.168.134.129: icmp_seq=1 ttl=64 time=1.68 ms  
64 bytes from 192.168.134.129: icmp_seq=2 ttl=64 time=0.519 ms  
64 bytes from 192.168.134.129: icmp_seq=3 ttl=64 time=0.874 ms  
64 bytes from 192.168.134.129: icmp_seq=4 ttl=64 time=0.489 ms  
--- 192.168.134.129 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3005ms  
rtt min/avg/max/mdev = 0.489/0.889/1.677/0.479 ms  
kali@kali: ~  
$
```

Figure 3. ping -c 4 192.168.134.129

Step 3. I performed a basic Nmap scan on the Metasploitable target to identify open ports and running services.



```
root@kali: ~  
$ nmap 192.168.134.129  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-02 11:56 EDT  
Nmap scan report for 192.168.134.129  
Host is up (0.002s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  mircrgrtry  
1324/tcp  open  ingreslock  
2000/tcp  open  nfs  
3123/tcp  open  cproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8080/tcp  open  http-alt  
8188/tcp  open  unknown  
MAC Address: 08:0C:29:1C:18:5B (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 16.63 seconds  
root@kali: ~  
$
```

Figure 4. nmap 192.168.134.129

Step 4. Service and Version Detection.

```
(root@kali)-[~]
└─$ nmap -sV 192.168.134.129
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-02 11:58 EDT
Nmap scan report for 192.168.134.129
Host is up (0.002% latency).
Not shown: 777 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain        ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rshexec
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath gmiiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5980/tcp  open  vnc            VNC (protocol 3.3)
6080/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8080/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:1C:18:5B (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 29.09 seconds
```

Figure 5. *nmap -sV 192.168.134.129*

Final Result:

The authorized local lab was successfully set up using Kali Linux and Metasploitable 2. The target IP was identified as 192.168.134.129, and connectivity was successfully verified with 0% packet loss. Nmap scans identified multiple open ports and running services, while the -sV scan provided detailed service and version information.

Conclusion:

The authorized local lab was successfully created using Kali Linux and Metasploitable 2. The target IP and network connectivity were verified, and Nmap was used to identify open ports, services, and their versions. This practical helped me understand the basics of network scanning and service identification.